

JIEJIA LIU

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PERSONAL INFORMATION

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Research interests:

Diffuse X-ray emission, Superbubbles, High-resolution spectroscopy, Instrumentation, Galaxy ecosystems.

EDUCATION

Ph.D. in Astronomy & Astrophysics
Department of Astronomy, Tsinghua University
Advisor: Prof. Wei Cui

2021 – expected December 2026

Bachelor of Physics
School of Physics, Nankai University

2017 – 2021

PUBLICATIONS

First Author

1. **Jiejia Liu**, Rui Huang, Chengzhe Li, et al., “X-rays from a newly discovered superbubble in M31”, 2024, [MNRAS](#), **533**, 3027.
2. **Jiejia Liu**, Sifan Wang, Hai Jin, et al., “Preliminary design of detector assembly for DIXE”, 2024, [JLTP](#), **216**, 273, [arXiv:2405.07559](#).
3. **Jiejia Liu**, Chunyang Jiang, Junjie Mao, et al., “Diffuse X-ray Explorer (DIXE): Sky survey strategy and collimator response demodulation”, 2026, [RAA](#), **26**, 125020, [arXiv:2606.10712](#).
4. **Jiejia Liu**, Junjie Mao, Shuinai Zhang, et al., “X-ray measurements of the elemental abundances for the diffuse emission in the nuclear starburst galaxy NGC 3079”, 2026, A&A accepted, [arXiv:2609.37646](#).
5. **Jiejia Liu**, Junjie Mao, Lei Sun, “In-flight validation of the wavelength-offset scaling relation for the XMM-Newton Reflection Grating Spectrometer”, in preparation.

Co-Author

1. Hai Jin, Junjie Mao, . . . , **Jiejia Liu**, et al., “Diffuse X-ray Explorer: A high-resolution X-ray spectroscopic sky surveyor on the China Space Station”, 2024, [JLTP](#), **215**, 256, [arXiv:2406.09813](#).
2. Ruixuan Tian, Junjie Mao, **Jiejia Liu**, et al., “Simulation of non X-ray background for the DIffuse X-ray Explorer (DIXE) mission”, 2026, [Exp. Astron.](#), **61**, 17, [arXiv:2604.13569](#).
3. Yu Zhou, **Jiejia Liu**, Volodymyr Takhistov, Kazuhisa Mitsuda, “Probing ALP-Photon Mixing with High-Resolution X-ray Spectroscopy”, ApJ accepted, [arXiv:2507.22006](#).
4. Michael C. H. Yeung, Martin G. F. Mayer, . . . , **Jiejia Liu**, et al., “The SRG/eROSITA diffuse soft X-ray background II. spectra and morphology of the eROSITA bubbles in the western Galactic hemisphere”, 2026, A&A accepted, [arXiv:2605.02998](#).
5. Jiayi Chen, Haiwu Pan, . . . , **Jiejia Liu**, et al., “Modeling and in-flight characterization of the point spread function for the Einstein Probe Wide-field X-ray Telescope”, RAA accepted.

APPROVED PROPOSALS

Co-I, 300 ks on Einstein Probe Cycle 2 (ID: 0102) 2025
Title: The Largest Hot Superbubbles Ever Seen in a Low-Density Environment?
PI: Dr. Rui Huang

PRESENTATIONS

Talks

X-rays from Superbubble Candidates in M31 2023.12
The 2nd HUBS Workshop, Haikou, China
Exposure Map and Point-Spread Function of DIXE 2024.12
The 2nd ShenKuo Symposium, Xiamen, China
X-rays from the superbubble in M31: a candidate for positive stellar feedbacks 2025.03
Science of Diffuse Sources in the New Era, Nanjing, China
DIXE collimator response and algorithm implementation of DIXE imaging 2025.06
Surveying the Hot Baryons Within the Milky Way ISSI-BJ Forum, Beijing, China

Posters

X-rays from Superbubble Candidates in M31 2023.05
The 26th Guo Shoujing Conference, Huangshan, China
X-rays from Superbubble Candidates in M31 2023.12
"Resolving Galaxy Ecosystem Across All Scales", Hong Kong, China
Preliminary Design of Detector Assembly for DIXE 2023.06
The 20th International Conference on Low Temperature Detectors, Daejeon, South Korea
Simulation of Magnetic Shielding for DIXE 2024.06
SPIE Astronomical Telescopes + Instrumentation, Yokohama, Japan

TEACHING EXPERIENCES

Teaching Assistant:
Major course, *Student Seminar* 2023
Major course, *Radiative Process in Astrophysics* 2024
Elective course, *Frontiers of Astrophysics* 2025

SCHOLARSHIPS & AWARDS

- Comprehensive Scholarship, First Prize, Tsinghua University 2025
- Comprehensive Scholarship, Second Prize, Tsinghua University 2022, 2024
- AMD Scholarship, Tsinghua University 2023
- "Gong Neng" Scholarship, Nankai University 2020
- Academic Excellence Scholarship, Nankai University 2018, 2019

REFERENCES

Prof. Wei Cui at Tsinghua University

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Prof. Junjie Mao at Tsinghua University

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Prof. Zhijie Qu at Tsinghua University

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Dr. Yu Zhou at National Astronomical Observatories

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